Seminar $\mathbb{N}_2 - 8$ 23.11.2016.

Recapitulare

① $V_1 = Sp\left\{\begin{pmatrix}1\\2\\0\end{pmatrix}, \begin{pmatrix}2\\1\\2\end{pmatrix}, \begin{pmatrix}3\\3\\2\end{pmatrix}\right\}$ B_1 bază pentru V_1

$V_2 = \left\{\begin{pmatrix}x\\y\\z\end{pmatrix} \mid \in \mathbb{R}^3 / 3x+y-2z=0,\ 3x-2y+z=0\right\}$ B_2 bază pentru V_2

$\dim_{\mathbb{R}} Sp(B_1 \cup B_2) = ?$

② $f: \mathbb{R}^3 \to \mathbb{R}^3$, $f\begin{pmatrix}x\\y\\z\end{pmatrix} = \begin{pmatrix}\frac{2}{9}(2x-y-2z)\\ -\frac{1}{9}(2x-y-2z)\\ -\frac{2}{9}(2x-y-2z)\end{pmatrix}$

a) $f \circ f = f$

b) bază, dim pentru $\ker f$, $\operatorname{Im} f$, $\ker f \cap \operatorname{Im} f$

③ $f: \mathbb{R}_{\le 2}[X] \to \mathbb{R}^4$, $f(P(x)) = \begin{pmatrix}P(1)\\ P(0)+P(1)\\ 2P(0)\\ \frac{1}{2}P''(0)\end{pmatrix}$

a) $A = M(f)$ în raport cu bazele canonice QR

b) Pseudosoluția sistemului $A\begin{pmatrix}x\\y\\z\end{pmatrix} = \begin{pmatrix}2\\0\\1\end{pmatrix}$

④ $A = \begin{pmatrix}0 & 0 & -2\\ 1 & 2 & 1\\ 1 & 0 & 3\end{pmatrix}$

$\lim_{m \to \infty} tr_{\mathbb{R}}(A^m) = ?$

⑤ $A = \begin{pmatrix}1 & 4\\ 2 & 8\end{pmatrix}$

a) SVD

b) $proj_{col(A)}\begin{pmatrix}1\\0\end{pmatrix} = ?$

c) Determinați pseudosoluția sistemului $A\begin{pmatrix}x\\y\end{pmatrix} = \begin{pmatrix}1\\0\end{pmatrix}$

① Observăm că $v_3 = v_2 + v_1 \Rightarrow v_1, v_2, v_3$ - liniar dependenți

Dar v_1, v_2 - liniar independenți $\Rightarrow v_1, v_2$ - bază în $V_1 \Rightarrow \dim V_1 = 2$

$$\begin{cases} 3x + y - 2z = 0 \\ 3x - 2y + z = 0 \end{cases}$$

$3y - 3z = 0 \Rightarrow y = z$

$3x + y - 2y = 0 \Rightarrow 3x - y = 0 \Rightarrow x = \frac{1}{3}y$

$V_2 = \left\{ \begin{pmatrix} y/3 \\ y \\ y \end{pmatrix} \mid y \in \mathbb{R} \right\} = Sp \begin{pmatrix} 1/3 \\ 1 \\ 1 \end{pmatrix} \Rightarrow \dim V_2 = 1$

$\#B_1 = 2$

$\#B_2 = 1$

$\#B_1 \cup B_2 \in \{2, 3\}$

$V_1 \cap V_2 = 0$

Fie v din intersecție $\Rightarrow v = \begin{pmatrix} y/3 \\ y \\ y \end{pmatrix} = \lambda_1 \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + \lambda_2 \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$

$$\begin{vmatrix} 1/3 & 1 & 2 \\ 1 & 2 & 1 \\ 1 & 0 & 2 \end{vmatrix} = \frac{4}{3} + 1 - 4 - 2 = \frac{4}{3} - 5 \neq 0 \Rightarrow \lambda_1 = \lambda_2 = 0$$

$\Rightarrow Sp(B_1 \cup B_2) = \mathbb{R}^3$

②

$1, X, X^2, \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}_{e_1} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}_{e_2} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}_{e_3} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}_{e_4}$

$f(1) = \begin{pmatrix} 1 \\ 2 \\ 2 \\ 0 \end{pmatrix}$

$f(X) = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}$

$f(X^2) = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 1 \end{pmatrix}$

$$A=\begin{pmatrix}1&1&1\\1&1&1\\2&0&0\\0&0&1\end{pmatrix}\quad \underset{u_1}{v_1}\ \underset{u_2}{v_2}\ \underset{u_3}{v_3}$$

$$u_1=v_1$$

$$u_2=v_2-\underset{u_1}{\text{proj}}\,v_3=\begin{pmatrix}1\\1\\0\\0\end{pmatrix}-\frac{3}{3}\begin{pmatrix}1\\1\\0\\0\end{pmatrix}=\begin{pmatrix}1\\1\\0\\0\end{pmatrix}-\begin{pmatrix}1\\1\\0\\0\end{pmatrix}=\begin{pmatrix}0\\0\\0\\0\end{pmatrix}$$

$$u_3=v_3-\underset{u_1}{\text{proj}}\,v_3-\underset{u_2}{\text{proj}}\,v_3=\begin{pmatrix}1\\1\\0\\1\end{pmatrix}-\frac{3}{3}\begin{pmatrix}1\\1\\0\\0\end{pmatrix}-\frac{1}{1}\begin{pmatrix}0\\0\\0\\1\end{pmatrix}=\begin{pmatrix}0\\0\\0\\0\end{pmatrix}$$

$$q_1=\frac{1}{\|u_1\|}\cdot u_1=\frac{1}{3}\begin{pmatrix}1\\1\\2\\0\end{pmatrix}=\begin{pmatrix}1/3\\1/3\\2/3\\0\end{pmatrix}$$

$$q_2=\frac{1}{\|u_2\|}\,u_2=\frac{1}{1}\begin{pmatrix}0\\0\\0\\0\end{pmatrix}$$

$$q_3=\frac{1}{\|u_3\|}\cdot u_3=\frac{1}{1}\begin{pmatrix}0\\0\\0\\1\end{pmatrix}$$

$$Q=(q_1\,|\,q_2\,|\,q_3)=\begin{pmatrix}1/3&2/3&0\\2/3&1/3&0\\2/3&-2/3&0\\0&0&1\end{pmatrix}$$

$$v_1=(v_1\cdot q_1)\,q_1=3q_1$$

$$v_2=(v_2\cdot q_1)\cdot q_1+q_2(v_2\cdot q_2)$$

$$v_3=(v_3\cdot q_1)\cdot q_1+q_2(v_3\cdot q_2)+q_3(v_3\cdot q_3)$$

$$R=\begin{pmatrix}3&1&1\\0&1&1\\0&0&1\end{pmatrix}$$

$$A=Q\cdot R$$

$$A^t\cdot A\begin{pmatrix}x\\y\\z\end{pmatrix}=A^t\begin{pmatrix}1\\2\\0\\1\end{pmatrix}$$

II $A = QR$

$A^t = R^t \cdot Q^t$

$R^t \cdot \underbrace{Q^t \cdot Q}_{I_3} \cdot R \begin{pmatrix} x \\ y \\ z \end{pmatrix} = R^t \cdot Q^t \begin{pmatrix} 1 \\ 2 \\ 0 \\ 1 \end{pmatrix} \quad / \cdot (R^t)^{-1}$

$R \begin{pmatrix} x \\ y \\ z \end{pmatrix} = Q^t \begin{pmatrix} 1 \\ 2 \\ 0 \\ 1 \end{pmatrix} \Rightarrow \begin{cases} 3x^* + y^* + z^* = 5/3 \\ y^* + 3z^* = 4/3 \\ z^* = 1 \end{cases} \Rightarrow x^* = \frac{1}{3}$

$\Rightarrow y^* = \frac{4}{3} - 1 = \frac{1}{3}$

$Q^t \begin{pmatrix} 1 \\ 2 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1/3 & 2/3 & 2/3 & 0 \\ 2/3 & 1/3 & -2/3 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 5/3 \\ 4/3 \\ 1 \end{pmatrix}$

④

$A = \begin{pmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{pmatrix}$

$\begin{vmatrix} -\lambda & 0 & 2 \\ 1 & 2-\lambda & 1 \\ 1 & 0 & 3-\lambda \end{vmatrix} \overset{L_1+L_2+L_3}{=\!=\!=} \begin{vmatrix} 2-\lambda & 2-\lambda & 2-\lambda \\ 1 & 2-\lambda & 1 \\ 1 & 0 & 3-\lambda \end{vmatrix} = (2-\lambda) \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2-\lambda & 1 \\ 1 & 0 & 3-\lambda \end{vmatrix} =$

$= (2-\lambda) \begin{vmatrix} 1 & 1 & 1 \\ 0 & 1-\lambda & 0 \\ 0 & -1 & 2-\lambda \end{vmatrix} = (2-\lambda)(1-\lambda)(2-\lambda) = (2-\lambda)^2(1-\lambda) = 0 \Rightarrow$

$\Rightarrow \begin{cases} \lambda_1 = 2 \\ \lambda_2 = 1 \end{cases} \Rightarrow \begin{cases} a_{\lambda_1} = 2 \\ a_{\lambda_2} = 1 \end{cases}$

I. $Av_1 = \lambda_1 v_1 \Rightarrow \begin{cases} -2z = 2x \\ x + 2y + z = 2y \\ x + 3z = 1z \end{cases} \Leftrightarrow \quad x = -z$

$V_{\lambda_1} = \left\{ \begin{pmatrix} x \\ y \\ -x \end{pmatrix} \mid x, y \in \mathbb{R} \right\} = \mathrm{Sp} \left(\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right) \Rightarrow g_{\lambda_1} = 2 = a_{\lambda_1}$

II. $AV_2 = \lambda_2 V_2$

$$\begin{cases} -2z = x \\ x + 2y + z = y \\ x + 3z = z \end{cases} \Rightarrow y - z = 0 \Rightarrow y = z$$

$$V_{\lambda_2} = \left\{ \begin{pmatrix} -2z \\ z \\ z \end{pmatrix} \mid z \in \mathbb{R} \right\} = Sp \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} \Rightarrow g_{\lambda_1} = 1 = a_{\lambda_1}$$

$A = P \cdot D \cdot P^{-1}$

$P = (V_1 | V_2 | V_3) = \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \\ -1 & 0 & 1 \end{pmatrix}$ ↳ vectori proprii

$D = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

$A^m = (P \cdot D \cdot P^{-1})^m = P \cdot D^m \cdot P^{-1}$

$D^m = \begin{pmatrix} 2^m & 0 & 0 \\ 0 & 2^m & 0 \\ 0 & 0 & 1 \end{pmatrix}$ $\Rightarrow A^m = P \cdot \begin{pmatrix} 2^m & 0 & 0 \\ 0 & 2^m & 0 \\ 0 & 0 & 1 \end{pmatrix} \cdot P^{-1}$

$tr(A^m) = tr(P \cdot D^m \cdot P^{-1}) = tr(P \cdot P^{-1} \cdot D^m) = tr(D^m) = 2^m \cdot 2 + 1$

$\boxed{tr(AB) = tr(BA)}$

$\boxed{tr(A+B) = tr A + tr B}$

② $f : \mathbb{R}^3 \to \mathbb{R}^3$, $f \begin{pmatrix} x \\ y \\ z \end{pmatrix} = (2x - y - 2z) \begin{pmatrix} 2/9 \\ -1/9 \\ -2/9 \end{pmatrix}$ $M(f) \cdot M(f)$

a) $f \circ f = f \Leftrightarrow M(f \circ f) = M(f)$

$f \circ f \begin{pmatrix} x \\ y \\ z \end{pmatrix} = f\left(f \begin{pmatrix} x \\ y \\ z \end{pmatrix} \right) = f \begin{pmatrix} \frac{2}{9}(2x - y - 2z) \\ \frac{-1}{9}(2x - y - 2z) \\ \frac{-2}{9}(2x - y - 2z) \end{pmatrix} =$

$$= \begin{pmatrix} \frac{2}{9}\left(2\cdot\frac{2}{9}(2x-y-2z)-\left(-\frac{1}{9}\right)(2x-y-2z)-2\cdot\left(-\frac{2}{9}\right)(2x-y-2z)\right) \\ - \quad - \quad - \\ - \quad - \quad - \end{pmatrix}$$

$$M(f) = \begin{pmatrix} 4/9 & -2/9 & -4/9 \\ -2/9 & 1/9 & 2/9 \\ -4/9 & 2/9 & 4/9 \end{pmatrix}$$

Fie M(f) matricea asociata lui f in raport cu o baza (de exemplu, baza canonica). Atunci f o f = f daca si numai daca M(f)M(f)=M(f)

$$f\begin{pmatrix}1\\0\\0\end{pmatrix} = 2\begin{pmatrix}2/9\\-1/9\\-2/9\end{pmatrix}$$

$$f\begin{pmatrix}0\\1\\0\end{pmatrix} = -\left(\quad\right)$$

$$f\begin{pmatrix}0\\0\\1\end{pmatrix} = -2\left(\quad\right)$$

b) $\ker f$, $\operatorname{Im} f$, $\ker f \cap \operatorname{Im} f$

$\ker f$

$$\begin{pmatrix}x\\y\\z\end{pmatrix} \in \ker f \Leftrightarrow f\begin{pmatrix}x\\y\\z\end{pmatrix} = \begin{pmatrix}0\\0\\0\end{pmatrix} \Leftrightarrow (2x-y-2z)\begin{pmatrix}2/9\\-1/9\\2/9\end{pmatrix} =$$

$$\Leftrightarrow (2x-y-2z)=0 \Rightarrow \ker f = \left\{\begin{pmatrix}x\\y\\z\end{pmatrix} \mid 2x-y-2z=0\right\} =$$

$$= \left\{\begin{pmatrix}x\\2x-2z\\z\end{pmatrix} \mid x,z\in\mathbb{R}\right\} = Sp\left\{\begin{pmatrix}1\\2\\0\end{pmatrix}, \begin{pmatrix}0\\-2\\1\end{pmatrix}\right\}$$

$$\dim_{\mathbb{R}} \operatorname{Im} f = 3 - \dim_{\mathbb{R}} \ker f = 1$$

$$b = \left\{\begin{pmatrix}2/9\\-1/9\\-2/9\end{pmatrix}\right\} \rightarrow \text{Baze in } \operatorname{Im} f$$

$$B = \left\{\begin{pmatrix}2\\-1\\-2\end{pmatrix}\right\}$$

Verificăm dacă $\begin{pmatrix} 2 \\ -1 \\ -2 \end{pmatrix} \in \operatorname{Ker} f$

$2 \cdot 2 - (-1) - 2(-2) = 9 \neq 0 \Rightarrow \begin{pmatrix} 2 \\ -1 \\ -2 \end{pmatrix} \notin \operatorname{Ker} f$

$\operatorname{Ker} f \cap \operatorname{Im} f = \{0\}$

$\boxed{\dim_{\mathbb{R}} \{0\} = 0}$

(5)

e) $A = \begin{pmatrix} 1 & 4 \\ 2 & 8 \end{pmatrix} = U \cdot \Sigma \cdot V^t$
$\quad\quad\quad

$v_2 = \frac{1}{\sqrt{17}} \begin{pmatrix} -4 \\ 1 \end{pmatrix}$

$V = (v_1 | v_2) = \begin{pmatrix} \frac{1}{\sqrt{17}} & -\frac{4}{\sqrt{17}} \\ \frac{4}{\sqrt{17}} & \frac{1}{\sqrt{17}} \end{pmatrix}$

Pas 2:

$u_1 = \frac{1}{\|v_1\|} A \cdot v_1 = \frac{1}{\sqrt{85}} \begin{pmatrix} 1 & 4 \\ 2 & 8 \end{pmatrix} \frac{1}{\sqrt{17}} \begin{pmatrix} 1 \\ 4 \end{pmatrix} = \frac{1}{17\sqrt{5}} \begin{pmatrix} 17 \\ 34 \end{pmatrix} = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

Completez pe u_1 la o bază ortonormată în $\mathbb{R}^2$.

Caut u_2 e.t. $\begin{cases} u_1 \cdot u_2 = 0 \\ \|u_2\| = 1 \end{cases}$

$u_2 = \begin{pmatrix} x \\ y \end{pmatrix}$

$u_1 \cdot u_2 = \frac{1}{\sqrt{5}} (x + 2y) = 0$

$x = -2y$

$u_2 = \begin{pmatrix} -2y \\ y \end{pmatrix} = y \begin{pmatrix} -2 \\ 1 \end{pmatrix}$

$\|u_2\| = |y| \cdot \sqrt{5}$

$y = \frac{1}{\sqrt{5}} \Rightarrow u_2 = \frac{1}{\sqrt{5}} \begin{pmatrix} -2 \\ 1 \end{pmatrix}$

$U = (u_1 | u_2) = \begin{pmatrix} \frac{1}{\sqrt{5}} & -\frac{2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{pmatrix}$

v_2 $\quad$ u_1

$\ker(A)$ $\quad$ $\mathrm{Col}(A)$

$\perp$ $\quad$ $\perp$

$\mathrm{Col}(A^t)$ $\quad$ $\ker(A^t)$

v_1 $\quad$ u_2

b) $\mathrm{proj}_{\mathrm{Col}(A)} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \left(\frac{\begin{pmatrix} 1 \\ 0 \end{pmatrix} \cdot u_1}{u_1 \cdot u_1} \right) u_1 = \frac{\frac{1}{\sqrt{5}}}{1} \cdot \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1/5 \\ 2/5 \end{pmatrix}$

$u_1 = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

c) $A \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

(5)

$$\begin{cases} x+4y=1 \\ 2x+8y=0 \end{cases}$$

$$A\begin{pmatrix} x \\ y \end{pmatrix} = \underset{\text{Col}(A)}{\text{proj}} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1/5 \\ 2/5 \end{pmatrix} \Rightarrow \begin{cases} x+4y=1/5 \\ 2x+8y=2/5 \end{cases} \Rightarrow \begin{pmatrix} \frac{1}{5}-4y \\ y \end{pmatrix} / y \in \mathbb{R}$$

$$\left\| \begin{pmatrix} \frac{1}{5}-4y \\ y \end{pmatrix} \right\|^2 = \left(\frac{1}{5}-4y\right)^2 + y^2 \Rightarrow (9)$$